# Example longform document

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# Contents

**1 A Sample Chapter** **2**
1.1 Structured statements . . . . . 2
1.2 Diagrams and ordinary lists . . . . . 2

**2 Another Chapter** **4**
2.1 Resetting theorem numbers . . . . . 4

# Chapter 1

# A Sample Chapter

## 1.1 Structured statements

**Definition 1.1.1.** A *longform object* is an object $X$ equipped with a map

$$\alpha_X : X \to G(X)$$

satisfying the following properties. ⚠ Here is a draft note.

(1.1.1.1) The object $X$ exists in the category under consideration.

(1.1.1.2) It is unique up to unique isomorphism.

**Proposition 1.1.2.** *Every longform object is determined by the data in Items (1.1.1.1) and (1.1.1.2).*

*Proof.* This is a template proof. The point is that the item references are attached to the theorem-like environment Theorem 1.1.1, so their labels are numbered using the surrounding theorem number. □

*Remark* 1.1.3. Long-form documents often contain many references. **Alice:** Alice's comment here. **Bob:** Bob's reply here. This example cites Hartshorne [Har77] and the Stacks Project [SP].

## 1.2 Diagrams and ordinary lists

The template supports commutative diagrams. For instance, one may write

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ C & \xrightarrow[\psi]{} & D. \end{array}$$

Such diagrams are useful in lecture notes and book-length projects.

The template also supports ordinary enumerated lists:

(a) A first ordinary item.

(b) A second ordinary item.

We can refer to Item (b) as an ordinary item, and to Item (1.1.1.2) as a theorem-numbered item.

# Chapter 2

# Another Chapter

## 2.1 Resetting theorem numbers

This chapter checks that theorem-like numbering behaves correctly after a chapter change.

**Lemma 2.1.1.** *The references from the first chapter, such as Theorem 1.1.1, still work from later chapters.*

*Proof.* This proof is only illustrative. It also demonstrates a displayed equation:

$$(f \circ g)^* = g^* \circ f^*. \tag{2.1.1}$$

□

*Example* 2.1.2. A long-form document may also contain external links, for instance to the biblatex package page.

# Bibliography

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977.

[SP] The Stacks Project Authors. *The Stacks Project*. URL: https://stacks.math.columbia.edu.